

A REALLY GOOD TITLE

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1 Introduction

2 Methods

2.1 Experimental design and data collection

The experiment was designed and implemented via the PsiTurk platform (Gureckis et al., 2015), and coded in a combination of JavaScript (specifically, the JsPsych library; de Leeuw, 2015) and Python. The study took place over two one-hour sessions, separated by six to eight days (mean 6.7 days; median 6.9 days) to approximate the common time span between releases of subsequent television episodes. Participants ($N = 70$) were divided into two equal groups. In the first session, all participants viewed the 25-minute pilot episode of the FX series, *Atlanta*. Following the episode, participants were then given up to an hour to recall its events, in order, to the best of their abilities. Mirroring Chen et al. (2017), they were asked to try to speak for a minimum of ten minutes and told that they were free to go back and describe earlier events, should they realize they missed any. Finally, they were given up to five minutes to predict what they thought would happen in the next episode. In the second session, all participants were again given up to one hour to recall to recall the episode viewed the prior week. Group one then viewed the second episode of *Atlanta*, while group two viewed the pilot of *Arrested Development*. Both groups finished by recalling the newly watched episode in the same manner. Eleven subjects' data were discarded for various reasons prior to processing and analysis: five did not follow the provided instructions, two did not complete the task, two did not return for the second session, one dropped out due to illness, and one was unable to complete the experiment due to technical issues. The data analyzed comprise the remaining 59 participants (30 in group one; 29 in group two).

2.2 Stimulus annotation and audio transcription

The ability of Heusser et al. (2018a)'s approach to model the explicit content of a naturalistic experience relies on the availability of detailed, hand-generated annotations that characterize the experience's evolving content. To create the annotation corpus for each television episode, I used ANVIL (Kipp, 2001), along with a custom XML specification file to record the following features for every camera shot:

- A brief description of physical occurrences and characters' actions ("*Narrative details: external events*")
- A brief description of characters' emotional states and various subtextual implications ("*Narrative details: internal states*")
- A list of characters appearing on-screen
- A transcription of all dialogue
- The name of the character speaking
- The shot's setting

- Whether the shot took place indoors or outdoors
- The presence or absence of music

These features were chosen both to be consistent with and to improve upon prior work with a similar set of stimulus annotation generated by Chen et al. (2017). Analyses carried out on Chen et al.’s feature set in conjunction with Heusser et al.’s modeling approach revealed that certain annotated features (e.g., what text appears on screen) contributed little-to-no structure to the stimulus trajectory. Further, some features (e.g., name of the character in focus and of the character speaking) were found to be highly redundant with each other, while others (e.g., the camera angle) were rarely preserved during recall and thus contributed unnecessary noise to the model. I chose to separate descriptions of physical actions from internal emotions and subtextual implications both to acknowledge that not all content is presented equally explicitly, and to enable future investigations by myself or others into potential differences in prioritization of those features in processing and memory.

All participant audio was transcribed using the Google Cloud Speech-to-Text API (Halpern et al., 2016). A previous proof-of-concept study (Ziman et al., 2018) suggested that this particular automatic transcriber may be used as an adequate stand-in for manual transcription of verbal response data, and as this data set amounted to approximately 50 hours of audio, I elected to use this approach. One significant asset to the Google transcription API is the option to bias the speech decoder toward certain words or phrase. For each audio file to be transcribed, I seeded the decoder with the union of words present in the hand-generated annotations of the reference episode (removing English stop-words, non-alphabetic characters, plurals, and possessives). Because Ziman et al. (2018) analyzed the quality of the Google transcriber in the context of a simple list-learning study, it was prudent to verify the decoding algorithm’s ability to return sufficiently accurate transcripts of natural language. Prior to transcribing the full data set, I tested the accuracy of the transcriber on a subset of audio files from five randomly chosen participants. I both automatically and manually transcribed each file, and constructed topic trajectories from each transcript method *insert ref to section describing topic modeling procedure!!!!*. On average, the automatically and manually generated recall trajectories were similar across timepoints (Pearson’s $r=0.63$, $SEM=0.06$) and their similarities to the corresponding episode trajectory (a rough estimate of recall accuracy) differed by less than 2.5%.

2.3 Dynamic content modeling

Following Heusser et al. (2018a), I created a high-dimensional space for each episode, in which its explicit content and participants’ corresponding recalls and predictions of that content may be compared. Specifically, I used `Hypertools` (Heusser et al., 2018b) to train a (distinct) topic model (?) on overlapping sliding windows of hand-generated annotations for each presented episode. This topic modeling approach

2.4 Event segmentation, and video-recall mapping

3 Analyses to be employed and expected results

3.1 Comparing immediate and delayed recall

3.2 Measuring the effect of contextual familiarity on future memory success

3.3 Relating prediction ability to memory success

4 Implications

References

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